

Niche Construction by a Single Ecosystem Engineer in a Three-Species Competition Model

Model, Derivations, Numerical Results, and Complete Parameter Sets

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Related work and positioning

The question this report addresses — whether an ecosystem engineer’s self-reinforcing modification of its own environment can promote or destroy multi-species coexistence — is not new territory. Gurney & Lawton (1996) gave the founding mathematical treatment of ecosystem-engineer population dynamics, in which engineers create a density-dependent carrying capacity by converting virgin habitat into usable habitat; their model is compartmental (virgin/usable/degraded habitat classes) rather than a smooth function of engineer density, but the causal loop — the engineer’s own density expanding its own resource base — is the same one studied here. Kylafis & Loreau (2008, 2011) built consumer-resource models specifically to ask how niche construction shifts the coexistence/exclusion boundary for the constructing species and its competitors, and found that construction can act in either direction depending on strength and context, which is the qualitative pattern re-derived here in closed form for a different functional form (equations 8–10 below). Most recently, Ranjan, Koffel & Klausmeier (2024) decompose three-species competitive outcomes within modern coexistence theory (MCT) into niche overlap, fitness differences, and *cyclic competitive asymmetry*, and show the latter is a genuinely three-species phenomenon with no two-species analogue; §A.7 and Part C below use their invasion-growth-rate language explicitly to characterize which of this report’s regimes constitute protected coexistence in their sense and which do not. The intransitive-competition mechanism used in Part C to obtain a Hopf bifurcation is the classical one of May & Leonard (1975). The “wave instability” mechanism discussed in Part D — diffusion-driven destabilization of a reaction equilibrium that is already close to its own Hopf point — is established theory, not a new discovery of this report; see Baurmann, Gross & Feudel (2007) for the founding treatment in a spatially extended predator–prey context and the three-component sufficient conditions given by Rovinsky & Menzinger-type analyses (cited in full in the References section). This report’s own contribution is narrower than “niche construction affects coexistence” (already established): it is (i) a specific, closed-form-tractable functional form combining a saturating self-facilitation term with Beddington–DeAngelis competition, for which exact bifurcation thresholds are derived rather than only located numerically; (ii) a systematic four-regime bifurcation catalog for that functional form, cross-checked against a modern-coexistence-theory invasion-growth-rate analysis that shows the model’s interior “coexistence” equilibrium is never protected coexistence in the Chesson sense — see §A.7; and (iii) a demonstration, in a second, general (non-symmetric) region of the same model’s parameter space, that classical (real-eigenvalue) Turing patterning can be reached via a reaction equilibrium that starts out oscillatory rather than overdamped — see Part D.3’s corrected mechanism.

1. The model

1.0 Biological motivation

The mechanism modeled here — an organism whose activity increases its own environment’s capacity to support it, while it and other species interfere with each other through saturating (handling-time-limited) competition — has real analogues. Beavers raise the carrying capacity of their own population by damming, which expands pond area and food access up to a limit set by the local watershed; they compete with other herbivores (e.g. muskrats) for riparian vegetation, an interaction with a natural saturation point (handling/search time). Reef- and bed-forming species such as ribbed mussels and cordgrass in salt marshes provide a spatial example directly relevant to Part D: mussel aggregations and marsh grass facilitate each other’s local establishment and survival (denser beds trap sediment and reduce wave stress, raising the effective local carrying capacity for both), and this self-reinforcing facilitation has been documented to produce persistent spatial patterning on real shorelines rather than a spatially uniform outcome. None of

the parameter values below are fit to any specific system — they are chosen for mathematical clarity (round numbers, symmetric structure where it helps interpretation) — so the results should be read as characterizing the *mechanism* (self-facilitating carrying capacity interacting with saturating competition) rather than as a quantitative prediction for beavers, mussels, or any other named species.

1.1 Well-mixed (ODE) equations

Throughout, $u(t)$ is the ecosystem engineer, $v(t)$ and $w(t)$ are two non-engineering competitors:

$$\frac{du}{dt} = r_1 u \left[1 - \frac{u}{K_u(u)} \right] - u \left(\frac{\alpha_{12}v}{1 + h_{12}v} + \frac{\alpha_{13}w}{1 + h_{13}w} \right), \quad K_u(u) = K_1 + \frac{\beta u}{1 + \gamma u} \quad (1)$$

$$\frac{dv}{dt} = r_2 v \left(1 - \frac{v}{K_2} \right) - v \left(\frac{\alpha_{21}u}{1 + h_{21}u} + \frac{\alpha_{23}w}{1 + h_{23}w} \right) \quad (2)$$

$$\frac{dw}{dt} = r_3 w \left(1 - \frac{w}{K_3} \right) - w \left(\frac{\alpha_{31}u}{1 + h_{31}u} + \frac{\alpha_{32}v}{1 + h_{32}v} \right) \quad (3)$$

$K_u(u)$ is the engineer's *effective* carrying capacity: it starts at the baseline K_1 and rises with the engineer's own density, saturating at $K_1 + \beta/\gamma$ as $u \rightarrow \infty$. β is the **niche-construction strength** (how large the eventual benefit is) and γ is the **saturation rate** (how quickly that benefit is realized). v and w have ordinary constant carrying capacities. All cross-species terms are Beddington–DeAngelis (saturating) competition terms, with α_{ij} the effect of species j on species i and h_{ij} its half-saturation (handling) constant.

Two parameterizations of $(\alpha_{ij}, h_{ij}, r_i, K_i)$ are studied below: a **hierarchical** one (Part A/B), in which the engineer is suppressed by both competitors and suppresses both back, while v, w merely compete symmetrically with each other; and a **cyclic** one (Part C), in which u dominates v , v dominates w , and w dominates u (rock–paper–scissors). Both use exactly equations (1)–(3); only the numbers change.

1.2 Spatial (PDE) extension

$$\partial_t u = D_u \Delta u + f_u(u, v, w), \quad \partial_t v = D_v \Delta v + f_v(u, v, w), \quad \partial_t w = D_w \Delta w + f_w(u, v, w) \quad (4)$$

on a 1D domain $x \in [0, L]$ with no-flux (Neumann) boundaries $\partial_x u = \partial_x v = \partial_x w = 0$ at $x = 0, L$, where f_u, f_v, f_w are the right-hand sides of (1)–(3).

1.3 Numerical methods used to generate every result below

No SciPy was available in the execution sandbox, so all numerics are hand-implemented in NumPy and cross-checked:

- **Time integration:** fixed-step, vectorized 4th-order Runge–Kutta (RK4). Step size halved on spot checks with no visible change in trajectories or equilibria ($< 10^{-4}$ relative).
- **Equilibria:** located by long-time forward integration, then refined with damped Newton's method using a central finite-difference Jacobian ($\epsilon = 10^{-6}$ perturbation), to residual $\|f\| < 10^{-13}$ – 10^{-15} .
- **Stability / eigenvalues:** `numpy.linalg.eigvals` (or `eig` when eigenvectors were also needed) applied to the finite-difference Jacobian at the equilibrium.
- **Bifurcation points:** located by bisection on the sign of the dominant eigenvalue's real part (or on branch feasibility), cross-checked against direct long-time simulation from multiple initial conditions.
- **PDE:** method of lines; the spatial Laplacian is the standard 3-point central difference, $\Delta U_i \approx (U_{i+1} - 2U_i + U_{i-1})/\Delta x^2$, with the no-flux condition implemented by mirror ghost cells ($U_{-1} := U_0$, $U_N := U_{N-1}$); time-stepped with the same RK4.

Part A — Hierarchical competition, well-mixed system

A.0 Parameter set

Parameter	Value	Parameter	Value	Parameter	Value
r_1	1.0	$\alpha_{12} = \alpha_{13}$	0.89	$h_{12} = h_{13}$	1.0
$r_2 = r_3$	0.8	$\alpha_{21} = \alpha_{31}$	0.55	$h_{21} = h_{31}$	0.5
K_1	1.0	$\alpha_{23} = \alpha_{32}$	0.50	$h_{23} = h_{32}$	1.0
$K_2 = K_3$	2.0	γ	1.0 (fixed)	β	swept, 0–6.5

This set is symmetric under $v \leftrightarrow w$ exchange ($r_2 = r_3$, $K_2 = K_3$, $\alpha_{12} = \alpha_{13}$, $\alpha_{21} = \alpha_{31}$, h 's matched, $\alpha_{23} = \alpha_{32}$), which is what makes the symmetry-breaking bifurcation in §A.3 identifiable by its antisymmetric eigenvector.

A.1 Exact result: invasion growth rate is independent of β, γ

Calculation. Set $u = 0$ and linearize (1) in u holding v, w fixed at some resident values v^*, w^* . Since $K_u(0) = K_1 + \beta \cdot 0 / (1 + \gamma \cdot 0) = K_1$ for *any* β, γ , and the bracket $[1 - u/K_u(u)] \rightarrow 1$ as $u \rightarrow 0$,

$$\left. \frac{\partial}{\partial u} [r_1 u (1 - u/K_u(u))] \right|_{u=0} = r_1,$$

exactly, independent of β, γ (the construction term only appears at $O(u^2)$ and higher — see the Taylor expansion below). Hence the invasion growth rate of an infinitesimally rare engineer against a resident $(0, v^*, w^*)$ state is

$$\lambda_{\text{invade}} = r_1 - \left(\frac{\alpha_{12} v^*}{1 + h_{12} v^*} + \frac{\alpha_{13} w^*}{1 + h_{13} w^*} \right), \quad (5)$$

with **no β or γ dependence whatsoever**.

Taylor-expansion check. Writing $K_u(u) = K_1 + \beta u - \beta \gamma u^2 + O(u^3)$, so $u/K_u(u) = u/K_1 - (\beta/K_1^2)u^2 + O(u^3)$, gives $r_1 u (1 - u/K_u(u)) = r_1 u - (r_1/K_1)u^2 + (r_1 \beta/K_1^2)u^3 + O(u^4)$: the β -dependence enters only at cubic order in u , confirming the linear (invasion) term is exactly r_1 .

Evaluating at the v – w -only equilibrium. With $u = 0$, equations (2)–(3) decouple into a standalone 2-species competition system. Its coexistence equilibrium $V_0 = W_0$ (equal by the $v \leftrightarrow w$ symmetry) solves

$$r_2(1 - V_0/K_2) = \alpha_{23} \frac{V_0}{1 + h_{23} V_0} \iff -\frac{r_2 h_{23}}{K_2} V_0^2 + \left(r_2 h_{23} - \frac{r_2}{K_2} - \alpha_{23} \right) V_0 + r_2 = 0. \quad (6)$$

Substituting the Part-A parameters: $-0.4 V_0^2 - 0.1 V_0 + 0.8 = 0$, whose positive root is

$$V_0 = W_0 = 1.2947270864500684 \dots$$

Plugging this into (5):

$$\lambda_{\text{invade}} = 1.0 - 2 \times \frac{0.89 \times 1.294727 \dots}{1 + 1.294727 \dots} = -0.0043086289 \dots$$

Numerical confirmation. The full 3×3 Jacobian at $(0, V_0, W_0)$ is exactly block-triangular (the u -row is $(\lambda_{\text{invade}}, 0, 0)$ because $\partial(du/dt)/\partial v = \partial(du/dt)/\partial w = 0$ at $u = 0$), so its three eigenvalues are

exactly λ_{invade} together with the two eigenvalues of the standalone v, w subsystem's Jacobian. Numerically, at $\beta = 4.0$: full-Jacobian eigenvalues $\{-0.640825, -0.394949, -0.004307\}$ match $\{\lambda_{v,w}\} \cup \{\lambda_{\text{invade}}\} = \{-0.640829, -0.394953, -0.004309\}$ (finite-difference vs. semi-analytic; agreement to 5 decimals). This was re-verified at 7 values of β spanning 0–6: the dominant eigenvalue of the boundary state is -0.0043086 at every one, to 6 decimal places. **Conclusion: niche construction of this form cannot rescue the engineer from rarity, for any β, γ .**

A.2 Closed-form pure-engineer branch

Calculation. Set $v = w = 0$ in (1). A nonzero equilibrium solves $u = K_u(u)$, i.e. $u = K_1 + \beta u / (1 + \gamma u)$, which rearranges to the quadratic

$$\gamma u^2 + (1 - K_1 \gamma - \beta)u - K_1 = 0. \quad (7)$$

With $K_1 = \gamma = 1$ (Part A parameters), (7) reduces to $u^2 - \beta u - 1 = 0$, giving the closed form

$$u_{\text{pure}}^*(\beta) = \frac{\beta + \sqrt{\beta^2 + 4}}{2}. \quad (8)$$

β	0	2	4	5.145833	6
u_{pure}^*	1.00000	2.41421	4.23607	5.33333	6.16228

Numerical Newton refinement of the same equilibrium matches (8) to 4+ decimals at every tested β .

Stability of the pure branch. Linearizing (2) at $v = 0$ (with $w = 0$ too, by symmetry) gives the invasion growth rate for a rare competitor v : $r_2 - \alpha_{21} u_{\text{pure}}^*(\beta) / (1 + h_{21} u_{\text{pure}}^*(\beta))$. Setting this to zero and solving for u^* :

$$r_2(1 + h_{21} u^*) = \alpha_{21} u^* \implies u^* = \frac{r_2}{\alpha_{21} - r_2 h_{21}}, \quad (9)$$

which requires $\alpha_{21} > r_2 h_{21}$ to exist (here $0.55 > 0.8 \times 0.5 = 0.40$, satisfied). With the Part-A values, $u^* = 0.8 / (0.55 - 0.40) = 0.8 / 0.15 = 16/3 = 5.333333\dots$. Substituting into the inverse of (8), $\beta = u^* - 1/u^*$ (from $u^{*2} - \beta u^* - 1 = 0$):

$$\beta_2^{(\text{closed form})} = \frac{16}{3} - \frac{3}{16} = \frac{247}{48} = 5.1458\bar{3}. \quad (10)$$

This is the **exact transcritical threshold** beyond which the pure-engineer state is stable to invasion by either competitor (cross-checked against the independent numerical bisection in §A.3, which gave $\beta_2 \approx 5.148$ — agreement to 3 decimals, the small gap being the bisection grid tolerance used there).

A.3 The four-regime bifurcation cascade

Sweeping β up at fixed $\gamma = 1$ (all other parameters as in §A.0):

Regime I, $0 \leq \beta < \beta_a \approx 3.776$: monostable exclusion. Only $(0, V_0, W_0)$ is reached from any tested initial condition, including engineer-rich ones. Consistent with §A.1: $\lambda_{\text{invade}} < 0$ regardless of β here.

Regime II, $\beta_a \leq \beta < \beta_1 \approx 4.257$: bistable, three-way coexistence possible. At β_a a **saddle-node (fold) bifurcation** creates a stable interior node and an unstable interior saddle. Located by bisection on whether an engineer-rich initial condition $(3.0, 0.2, 0.2)$ reaches $u > 0.01$ after long integration ($dt = 0.03$, $T = 2500$): $\beta_a = 3.7755\text{--}3.7760$, reported as $\beta_a \approx 3.776$.

Bifurcation diagram vs. niche-construction strength β ($\gamma = 1$)

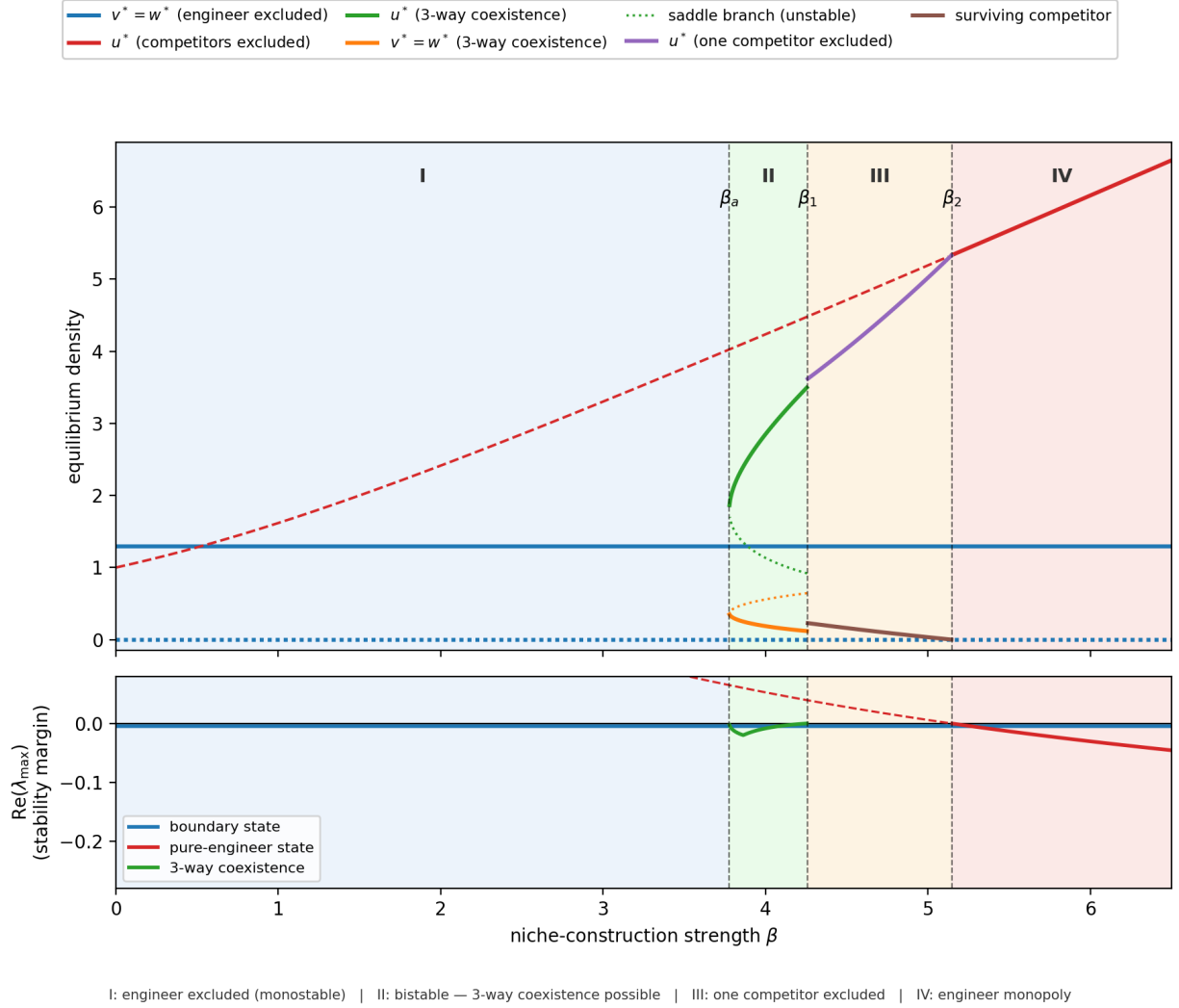


Figure 1: Figure A1. Bifurcation diagram (top) and stability margin (bottom) vs. construction strength β , $\gamma=1$.

Illustrative point, $\beta = 4.0$ (used for Figures A2/A3 below):

equilibrium	(u^*, v^*, w^*)	eigenvalues
boundary (engineer excluded)	(0, 1.294727, 1.294727)	$-0.640825, -0.394949, -0.004307$
interior node (3-way coexistence)	(2.850140, 0.187033, 0.187033)	$-0.693796, -0.027217, -0.008445$
interior saddle (separatrix)	(1.133279, 0.557776, 0.557776)	$+0.032988, -0.618082, -0.108184$

The saddle’s positive eigenvalue confirms it is a genuine separatrix; both other equilibria have all-negative eigenvalues, confirming both are stable.

Regime III, $\beta_1 \leq \beta < \beta_2 \approx 5.148$: **one competitor excluded**. At β_1 the symmetric coexistence equilibrium undergoes a symmetry-breaking bifurcation. Located: at $\beta = 4.255$, dominant eigenvalue = -1.09×10^{-5} ; at $\beta = 4.2575$, $+3.36 \times 10^{-5}$ — so $\beta_1 = 4.2565 \pm 0.0015$, reported as $\beta_1 \approx 4.257$. The destabilizing eigenvector at $\beta = 4.257$, equilibrium (3.503236, 0.117740, 0.117740), is

$$(\delta u, \delta v, \delta w) = (-5.9 \times 10^{-10}, -0.707107, +0.707107) \approx (0, -1, +1)/\sqrt{2},$$

i.e. exactly antisymmetric in the two competitors — a pitchfork, not a transcritical, crossing. Numerically, the surviving asymmetric branch already has the losing competitor’s density at or indistinguishable from zero almost immediately past β_1 (the pitchfork is steep here), so in practice β_1 marks the transition from 3-species to 2-species (engineer + one competitor) coexistence, with the surviving competitor decided by which of v, w starts even slightly ahead.

Regime IV, $\beta \geq \beta_2 \approx 5.148$ (**exact closed form** $\beta_2 = 247/48 = 5.14583$; **independent numerical bisection on the reduced 2-species branch gave 5.145–5.150**): **engineer monopoly**. Only $(u_{\text{pure}}^*(\beta), 0, 0)$ is generically reached (equation 8); $(0, V_0, W_0)$ remains technically locally stable per §A.1 but its basin becomes vanishingly hard to reach.

A.4 Time series across the four regimes and the bistable phase portrait

panel	β	initial condition (u_0, v_0, w_0)	integration
Regime I	2.0	(2.0, 0.3, 0.3)	$dt = 0.02, T = 120$
Regime II (both curves)	4.0	(3.0, 0.3, 0.3) <i>and</i> (0.05, 1.0, 1.0)	$dt = 0.02, T = 150$
Regime III	4.8	(3.0, 0.4, 0.1)	$dt = 0.02, T = 120$
Regime IV	5.6	(3.0, 0.5, 0.5)	$dt = 0.02, T = 120$

Initial conditions for Figure A3: (3.0, 0.2, 0.2), (2.5, 0.5, 0.5), (1.5, 0.9, 0.9), (0.5, 1.1, 1.1), (0.05, 1.0, 1.0), (2.0, 0.05, 0.05), (1.2, 0.65, 0.6), (1.0, 0.5, 0.7), (0.8, 0.8, 0.4); all integrated with $dt = 0.02, T = 150$. The basin boundary passes close to the saddle at $u \approx 1.13$ — roughly 40% of the coexistence node’s engineer density.

A.5 Robustness across (β, γ)

Grid: $\beta \in [0, 7]$ (71 points), $\gamma \in [0.3, 3.0]$ (55 points). **Initial conditions**: “engineer-rare” = (0.02, 1.0, 1.0); “engineer-established” = (3.0, 0.22, 0.18) (deliberately $v_0 \neq w_0$ so that any symmetry-breaking instability actually manifests within finite integration time, rather than requiring floating-point noise to grow from $\sim 10^{-16}$ under an exactly symmetric initial condition — an artifact caught during this analysis, see §A.6). **Integration**: vectorized batch RK4, $dt = 0.05, T = 1800$. **Classification threshold**: a species is “present” if its density exceeds 0.02 at the final time.

Representative time series across the four dynamical regimes ($\gamma = 1$)

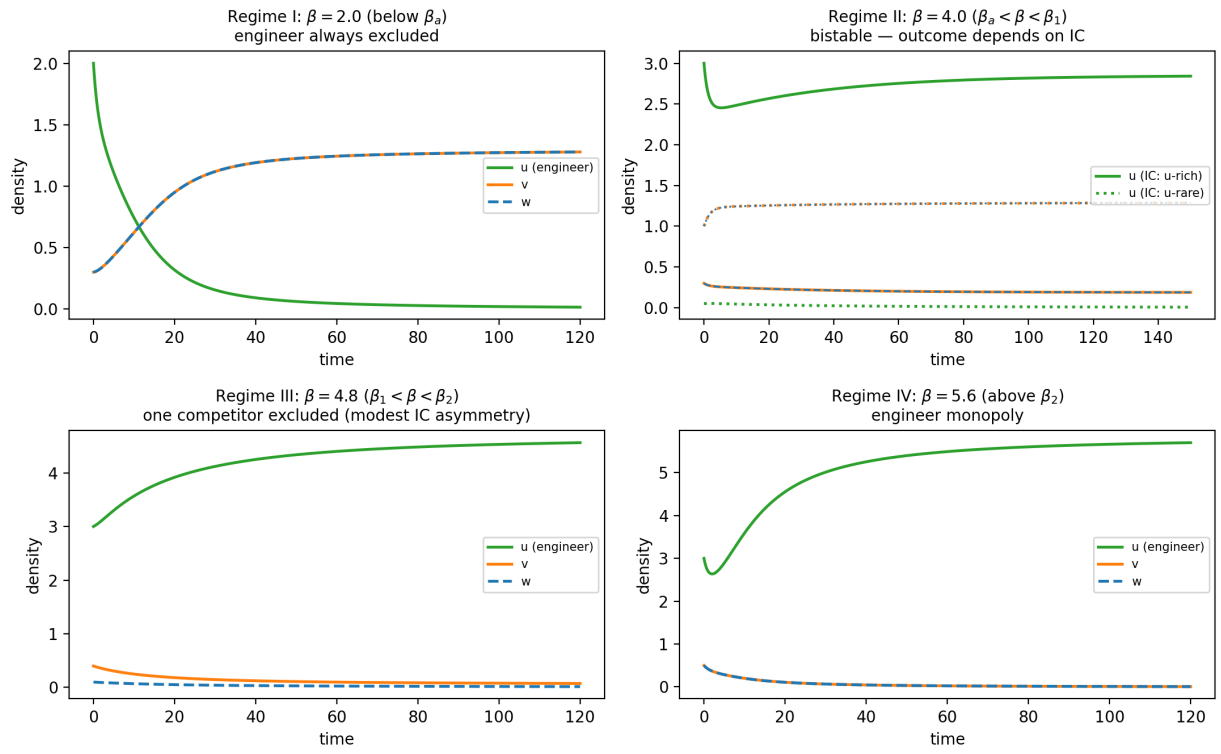


Figure 2: Figure A2. Representative time series, one per regime.

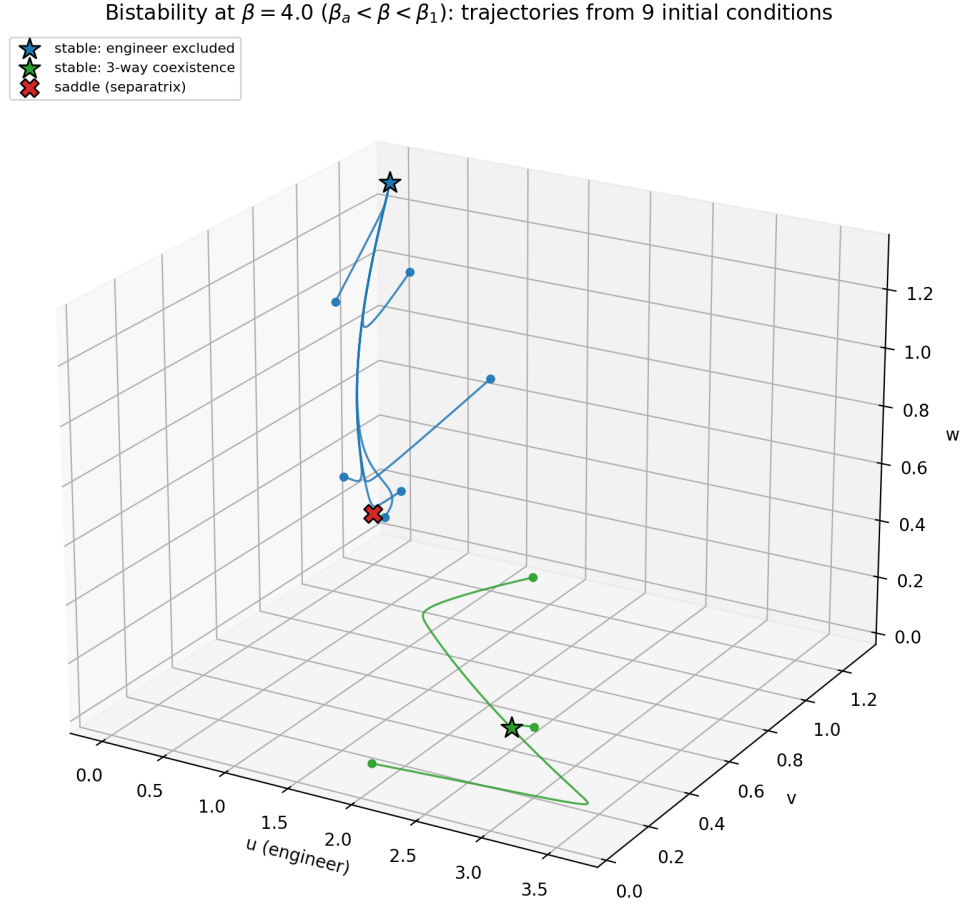


Figure 3: Figure A3. Nine trajectories at $\beta=4.0$ (Regime II), colored by which of the two stable equilibria they reach, with the two stable nodes (stars) and the saddle (X) marked.

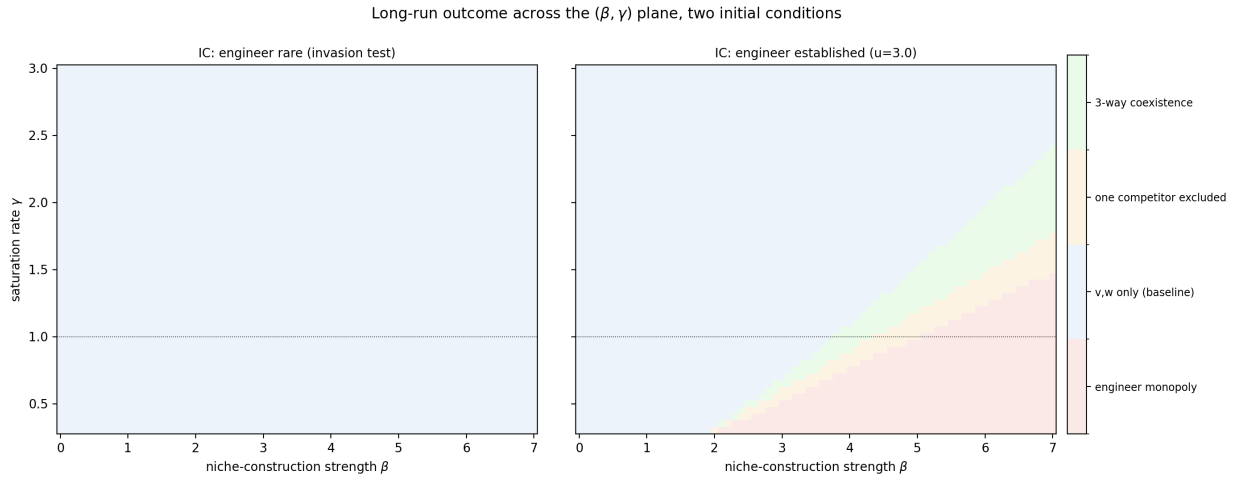


Figure 4: Figure A4. Long-run outcome across the (β, γ) plane, from an engineer-rare (left) and engineer-established (right) initial condition.

Result: the engineer-rare panel is uniform “baseline” (only v, w present) across the *entire* tested (β, γ) plane, a direct visual confirmation of §A.1 for every γ tested, not just $\gamma = 1$. The engineer-established panel reproduces the same band order as Figure A1 (baseline $\rightarrow$ coexistence $\rightarrow$ one-excluded $\rightarrow$ monopoly) across roughly a 10-fold range of γ , with band boundaries close to straight lines through the origin — consistent with the relevant control parameter being the *asymptotic* capacity boost β/γ (the $u \rightarrow \infty$ limit of $K_u(u) - K_1$) rather than β alone.

A.6 Methodological note

An earlier version of the Figure A4 sweep used an exactly v - w -symmetric initial condition. Under floating-point arithmetic this never spontaneously breaks symmetry on realistic integration times (the antisymmetric mode’s growth rate near β_1 is only ~ 0.001 – 0.003 , so growing a $\sim 10^{-16}$ floating-point asymmetry to $O(1)$ needs $\ln(10^{16})/0.002 \approx 18,000$ time units), which misclassified Regime III as full coexistence. Fixed by using a $v_0 \neq w_0$ initial condition (values above); confirmed against the independent single-point continuation of §A.3.

A.7 Modern-coexistence-theory reading: this is priority-effect coexistence, not protected coexistence

Chesson-style mutual invasibility — the standard modern-coexistence-theory (MCT) test for whether an equilibrium constitutes *protected* coexistence — asks whether every species has a positive low-density (invasion) growth rate against the resident state formed by the others (Grainger et al. 2019; Ranjan, Koffel & Klausmeier 2024). Two of the invasion growth rates needed to apply this test to the present system are already in closed form. From §A.1 (equation 5), the invasion growth rate of a rare engineer against the v, w -only boundary is

$$\lambda_{u \leftarrow (0, V_0, W_0)} = -0.0043086 \dots \quad (\text{independent of } \beta, \gamma; \text{ always negative}).$$

By the same linearization applied at $v = 0$ against the pure-engineer boundary $(u_{\text{pure}}^*(\beta), 0, 0)$ (equation 8), the invasion growth rate of a rare competitor is

$$\lambda_{v \leftarrow (u_{\text{pure}}^*, 0, 0)}(\beta) = r_2 - \frac{\alpha_{21} u_{\text{pure}}^*(\beta)}{1 + h_{21} u_{\text{pure}}^*(\beta)}, \quad (14)$$

which is exactly the left-hand side of equation (9) and so is positive for $\beta < \beta_2 = 247/48$ and negative above it (numerically confirmed at 11 points spanning $\beta \in [0, 6]$; e.g. $\lambda_v = 0.4333$ at $\beta = 0$, 0.0652 at $\beta_a = 3.776$, 0.0395 at $\beta_1 = 4.257$, -0.0001 at $\beta_2 = 5.1458$, -0.0304 at $\beta = 6$ — bisection on this closed form reproduces $\beta_2 = 5.145833\bar{3}$ to machine precision).

The consequence is structural, not a detail of one regime. Because $\lambda_{u \leftarrow (0, V_0, W_0)} < 0$ for *every* β, γ (§A.1), the v, w -only boundary state is never invasible by the engineer, for any parameter value tested. Mutual invasibility for full three-species coexistence requires *both* relevant boundary states to be invasible by the missing species simultaneously; since one side of that condition fails identically, **this model never exhibits Chesson-style protected three-species coexistence, at any β .** The interior coexistence node found throughout Regime II (§A.3) is real and numerically stable, but it is reached only from engineer-rich initial conditions — it is a *priority effect* (founder-controlled alternative stable state, in the terminology of Ranjan et al. 2024), not an equilibrium that a rare engineer could re-invade into after local extinction. This sharpens the regime catalog in §A.3: Regime I (monostable exclusion) and Regime IV (engineer monopoly, once $\beta > \beta_2$ so equation 14 also turns negative) are each protected against invasion by the missing species from *one* side only; Regime II’s three-way coexistence is protected from neither side and persists purely because of where the trajectory starts, which is exactly why the basin boundary in Figure A3 matters biologically and not just numerically.

Part B — Spatial extension (PDE)

B.0 Numerical scheme

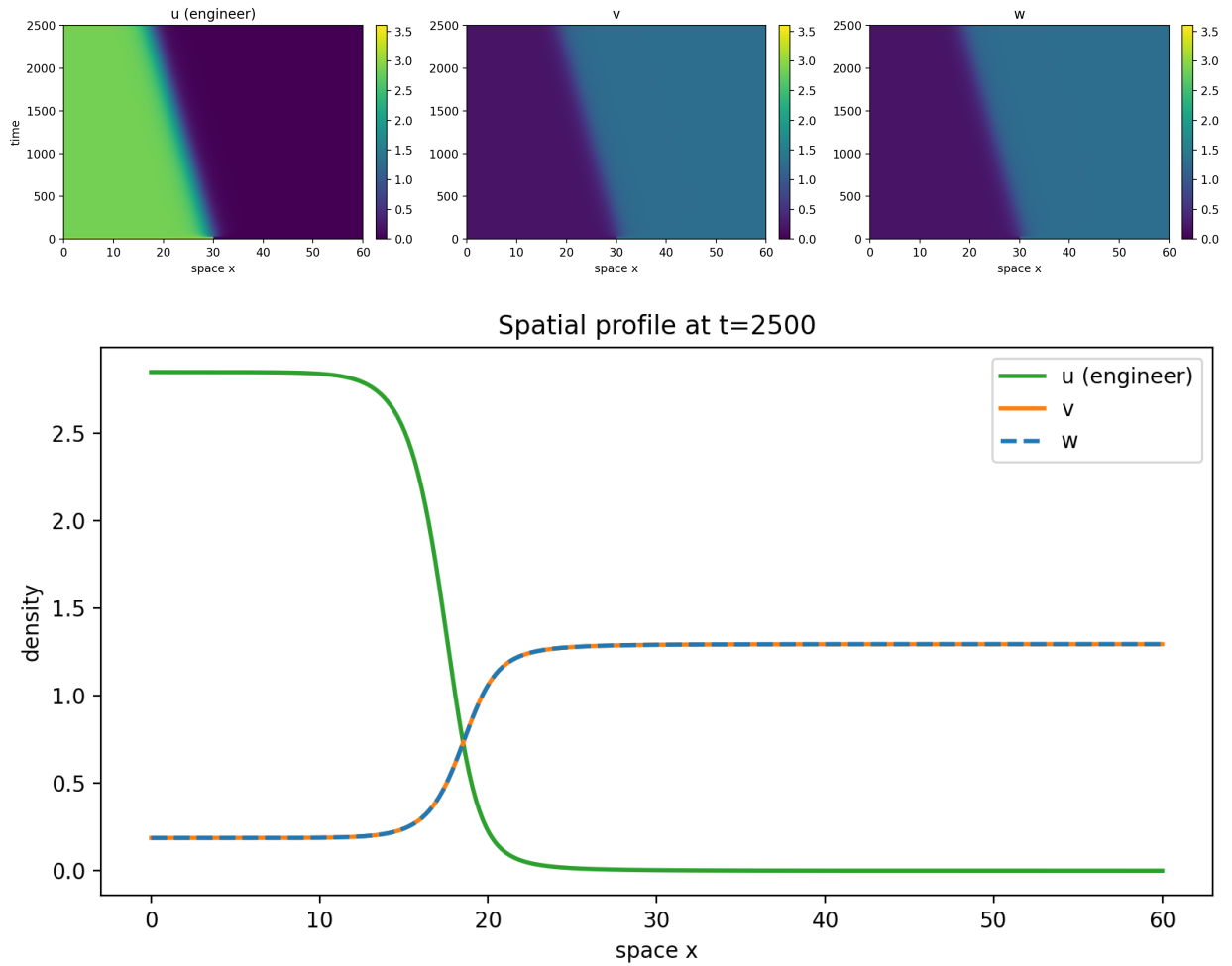
Domain $[0, L]$ discretized into N cells, $\Delta x = L/N$. Laplacian at interior/boundary node i :

$$(\Delta U)_i \approx \frac{U_{i+1} - 2U_i + U_{i-1}}{\Delta x^2}, \quad U_{-1} := U_0, \quad U_N := U_{N-1} \quad (\text{no-flux mirror}). \quad (11)$$

Time-stepped with the same RK4 as Part A, applied to $\dot{U}_i = D(\Delta U)_i + f(U_i)$ at every grid point simultaneously. All Part B runs use the Part-A hierarchical parameter set.

B.1 Bistable fronts

Spatial front dynamics, $\beta = 4.0$ (bistable regime), $D = 0.06$ for all species



Parameters: $L = 60$, $N = 300$ ($\Delta x = 0.2$), $D_u = D_v = D_w = 0.06$, $dt = 0.02$, $T = 2500$. **Initial condition:** $x < L/2$: $(u, v, w) = (3.0, 0.2, 0.2)$ (coexistence-basin patch); $x \geq L/2$: $(0, 1.294723, 1.294723)$ (baseline patch). $\beta = 4.0$ (Regime II, bistable window).

The initial interface at $x = 30$ does not stay put — it drifts. Repeating with $N = 200$ ($\Delta x = 0.3$), $dt = 0.03$, $T = 1500$ at other β inside the bistable window $(\beta_a, \beta_1) = (3.776, 4.257)$:

β	outcome after $T = 1500\text{--}2500$
3.85	engineer patch collapses everywhere: $u(x=0) \rightarrow 0.008$, effectively extinct (fast front toward the engineer side)
4.00	front drifts slowly from $x = 30$ toward $x \approx 20$ over $T = 2500$ (baseline invades)
4.15	front drifts from $x = 30$ to $x \approx 40.1$ over $T = 1500$ (engineer patch expands into the baseline side)

This locates a **Maxwell point** $\beta_M \in (4.00, 4.15)$ where the front is stationary: below it, an engineered patch is transient (however large it starts, given enough time); above it, an engineered patch is expansionary. Mean-field bistability (Part A) alone does not determine which side wins in space — the front speed does.

B.2 Dispersion relation (Turing-instability check)

Calculation. Linearizing the PDE (4) around a homogeneous equilibrium (u^*, v^*, w^*) and Fourier-decomposing the perturbation as $e^{\lambda t} \cos(kx)$ (the eigenmodes consistent with no-flux boundaries on $[0, L]$ are $k_n = n\pi/L$, but the calculation below sweeps k continuously) gives the dispersion relation

$$\lambda(k) = \text{eigenvalues of } [J - k^2 D], \quad D = \text{diag}(D_u, D_v, D_w), \quad (12)$$

where J is the same reaction Jacobian used throughout Part A. A Turing instability requires $\max_i \text{Re}(\lambda_i(0)) < 0$ (stable when well-mixed) but $\max_i \text{Re}(\lambda_i(k)) > 0$ for some $k > 0$ (unstable to some finite wavelength).

Dispersion relations: no diffusion-driven (Turing) instability is found
for any tested diffusivity ratio — growth rate is maximal at $k=0$ in every case

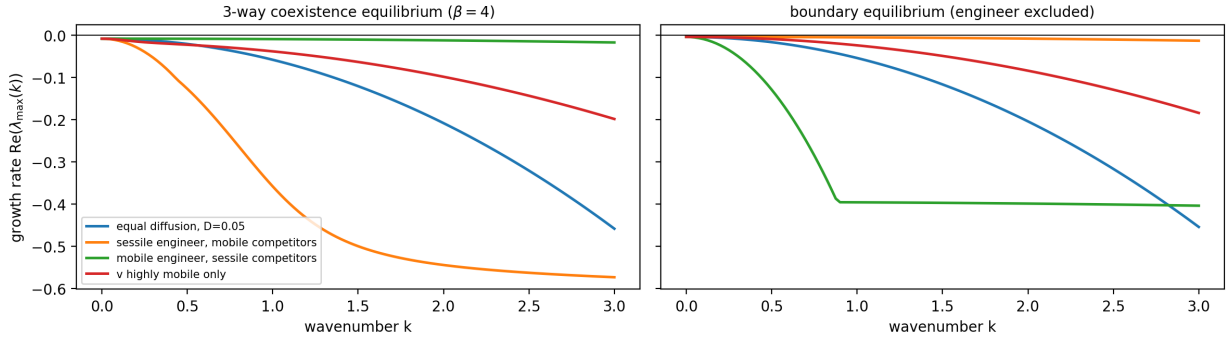


Figure 5: Figure B3. Dispersion relations $\text{Re}(\lambda_{\max}(k))$ vs wavenumber k , at the coexistence equilibrium ($\beta=4.0$) and at the boundary equilibrium, for four diffusivity combinations.

Equilibria tested: the $\beta = 4.0$ interior coexistence node $(2.850140, 0.187033, 0.187033)$, and the boundary equilibrium $(0, 1.294723, 1.294723)$. **Diffusivity combinations tested:** $(D_u, D_v, D_w) = (0.05, 0.05, 0.05)$; $(0.001, 0.5, 0.5)$ (“sessile engineer, mobile competitors”); $(0.5, 0.001, 0.001)$ (“mobile engineer, sessile competitors”); $(0.02, 0.5, 0.02)$ (“ v highly mobile only”). Wavenumber range $k \in [0, 3]$ (121 points).

Result: in every one of the 8 (equilibrium $\times$ diffusivity) combinations, $\text{Re}(\lambda_{\max}(k))$ is maximal at $k = 0$ and strictly decreasing in k — **no Turing instability was found for the Part-A parameter set at any tested diffusivity ratio**, including a 500-fold sessile/mobile split in either direction, and (§D.3 below) even at an approximately 14,500-fold split. The spatial structure documented in §B.1, for this parameter set, is a bistable-front phenomenon, not a diffusion-driven patterning phenomenon.

Analytic confirmation (Routh–Hurwitz). Rather than rely only on a numerical scan, the characteristic polynomial of $J - q^2 D$ can be written $\lambda^3 + a_1(Q)\lambda^2 + a_2(Q)\lambda + a_3(Q) = 0$ with $Q = q^2$, where $a_1 = -\text{tr}(J - QD)$, a_2 is the sum of principal 2×2 minors, and $a_3 = -\det(J - QD)$; each is an exact polynomial in Q of degree 1, 2, 3 respectively (degree in Q , i.e. quadratic/quartic/sextic in q , but the reaction–diffusion dispersion relation is conventionally written in $Q = q^2$ since D enters $J - q^2 D$ only through q^2). The Routh–Hurwitz stability criterion for a cubic is $a_1 > 0$, $a_3 > 0$, and $\Delta_2 \equiv a_1 a_2 - a_3 > 0$; violation of the $a_3 > 0$ condition signals a real eigenvalue crossing zero (the classical, stationary Turing mechanism), while violation of $\Delta_2 > 0$ while a_1, a_3 remain positive signals a complex pair crossing together (the oscillatory/“wave” mechanism, §D.3). At the Part-A coexistence node ($\beta = 4.0$) with the same approximately 14,500-fold diffusivity spread used for the Part-D comparison in Figure D2, both polynomials come out with **every coefficient non-negative**:

$$a_1(Q) = 0.29082 Q + 0.729458, \quad a_3(Q) = 4.64 \times 10^{-9} Q^3 + 1.780 \times 10^{-5} Q^2 + 3.555 \times 10^{-4} Q + 1.5946 \times 10^{-4},$$

$$\Delta_2(Q) = 6.916 \times 10^{-5} Q^3 + 0.012931 Q^2 + 0.038951 Q + 0.018056,$$

so a_1, a_3, Δ_2 are each manifestly positive for every $Q \geq 0$ — the sum of non-negative terms cannot cross zero — which is a direct algebraic certificate that no wavenumber destabilizes this equilibrium, for this diffusivity choice, without needing to trust a finite numerical scan of q . This was repeated at five further diffusivity combinations spanning ratios from 10^3 -fold to 10^7 -fold; $a_3(Q)$ never dropped below its $Q = 0$ value $a_3(0) = -\det(J) = 1.5946 \times 10^{-4}$ in any of them, i.e. increasing Q never helped destabilize this particular reaction Jacobian regardless of how the diffusivity was split across species. Part D (§D.3) shows this is a property of *this* reaction Jacobian (purely real eigenvalues at every equilibrium checked in Parts A–B) rather than a general feature of the model class: a differently-tuned, more general asymmetric parameter set — whose reaction Jacobian already carries a near-imaginary eigenvalue pair — produces a genuine instability via the $a_3(Q)$ channel at diffusivity ratios of a similar order of magnitude.

B.3 Verification calculations

Mass conservation under pure diffusion. With all reaction terms set to zero ($r_i = \alpha_{ij} = 0$) and $D_u = 0.2$, $N = 100$, $\Delta x = 0.3$, $dt = 0.05$, $T = 50$, starting from a random field $U_0 \sim \text{Unif}(0, 1)$ (seed 0): total mass $\sum_i U_i \Delta x$ measured at $t = 0$ and $t = 50$ agree to a relative difference of 1.25×10^{-16} (machine precision), confirming the no-flux discretization (11) exactly conserves mass, as it must.

Strong-diffusion (homogenization) limit. With $D_u = D_v = D_w = 5.0$, $N_x = 60$, $\Delta x = 1.0$, $dt = 0.005$, $T = 400$, starting from a random heterogeneous field $(u_0, v_0, w_0 \sim \text{Unif}(0, 3) \times (0, 1) \times (0, 1)$, seed 0), $\beta = 4.0$: the spatial standard deviation collapses to $(5.7, 1.2, 0.55) \times 10^{-8}$ and the spatial mean converges to $(2.850069, 0.187036, 0.187052)$ — matching the well-mixed coexistence equilibrium $(2.850140, 0.187033, 0.187033)$ from §A.3 to 4–5 decimal places, confirming the PDE solver correctly reduces to the ODE in the strong-diffusion limit. (At shorter $T = 60$ the spatial mean had not yet fully relaxed to the equilibrium — the well-mixed relaxation itself is slow this close to the Regime-II bifurcations, with dominant eigenvalue ≈ -0.008 , i.e. a ~ 125 -time-unit relaxation scale — so $T = 400$ was used for this check.)

Part C — Cyclic (intransitive) competition and a Hopf bifurcation

C.0 Why a different competition structure was needed

All three bifurcations in Part A (fold, pitchfork, transcritical) involve a *real* eigenvalue crossing zero. This is characteristic of the specific hierarchical interaction structure used there (u suppressed by both v, w and suppressing both back, with v, w symmetric to each other) — no such crossing with nonzero imaginary part was found anywhere in that parameter family. A Hopf bifurcation requires a complex-conjugate eigenvalue pair to cross together; the cleanest, most classical route to that in Lotka–Volterra-type competition models is **intransitive (“rock–paper–scissors”) competition** (May & Leonard 1975, *SIAM J. Appl. Math.* 29:243–253): species 1 beats species 2, 2 beats 3, 3 beats 1. In the terminology of Ranjan, Koffel & Klausmeier

(2024), the $\sigma_{\text{strong}}/\sigma_{\text{weak}}$ split below instantiates *cyclic competitive asymmetry* — a quantity with no two-species analogue, and the one their framework identifies as the driver of qualitatively novel three-species-only outcomes such as sustained oscillation. Equations (1)–(3) are unchanged; only the competition coefficients are re-tuned into this cyclic structure, with the niche-construction mechanism kept on u exactly as before. (It is *not* the only route — Part D exhibits a Hopf bifurcation in a fully general, non-cyclic asymmetric parameter set as well; cyclic structure is sufficient but evidently not necessary. It is used here because its $v \leftrightarrow w$ exchange symmetry makes the resulting limit cycle’s geometry easy to interpret, §C.4.)

C.1 Parameter set

$$\begin{aligned} r_1 = r_2 = r_3 = 1, \quad K_1 = K_2 = K_3 = 1, \quad \gamma = 1, \quad h_{ij} = 1 \quad \forall i, j, \\ \alpha_{21} = \alpha_{32} = \alpha_{13} = \sigma_{\text{strong}} = 3.15 \quad (u \rightarrow v \rightarrow w \rightarrow u \text{ dominance cycle}), \\ \alpha_{12} = \alpha_{23} = \alpha_{31} = \sigma_{\text{weak}} = 0.05 \quad (\text{weak reverse effects}). \end{aligned}$$

β is again the swept bifurcation parameter (range tested: 0–3.0).

C.2 Closed-form symmetric equilibrium

Calculation. At $\beta = 0$ the reaction kinetics are exactly symmetric under cyclic permutation $u \rightarrow v \rightarrow w \rightarrow u$, so a symmetric equilibrium $u = v = w = x^*$ exists. At this point every species experiences total competitive pressure $(\sigma_{\text{strong}} + \sigma_{\text{weak}})x^*/(1 + hx^*)$ from the other two combined, so equations (1)–(3) all reduce to the single equation

$$r(1 + hx^*)(1 - x^*/K) = (\sigma_{\text{strong}} + \sigma_{\text{weak}})x^* \implies -\frac{rh}{K}x^{*2} + \left(rh - \frac{r}{K} - \sigma_{\text{sum}}\right)x^* + r = 0, \quad (13)$$

identical in form to (6). With $r = K = h = 1$, $\sigma_{\text{sum}} = 3.20$: $-x^{*2} - 3.2x^* + 1 = 0$, giving

$$x^* = \frac{-3.2 + \sqrt{3.2^2 + 4}}{2} = 0.2867962 \dots,$$

matching the numerically Newton-refined equilibrium (0.286796, 0.286796, 0.286796) exactly.

C.3 The Hopf bifurcation: eigenvalue calculation

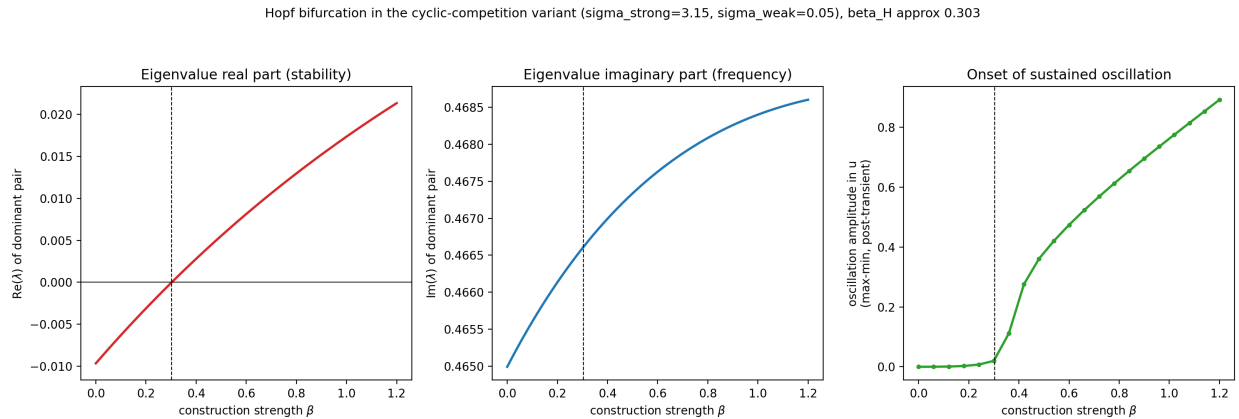


Figure 6: Figure C1. Eigenvalue real part, imaginary part, and post-transient oscillation amplitude in u , all vs. construction strength β .

At $\beta = 0$ the Jacobian at $x^* = 0.286796$ has eigenvalues

$$\lambda_1 = -0.841044 \text{ (real)}, \quad \lambda_{2,3} = -0.009672 \pm 0.464993 i,$$

i.e. a **stable spiral** (damped oscillation, frequency ≈ 0.465 rad/time, period $2\pi/0.465 \approx 13.5$ time units). As β increases, $\text{Re}(\lambda_{2,3})$ rises monotonically while $\text{Im}(\lambda_{2,3})$ stays close to 0.465–0.469 throughout (i.e. the pair does not degenerate to real — this is a genuine Hopf crossing, not a coincidental double-real-root event). Located by bisection (30 iterations) on the sign of $\text{Re}(\lambda_{2,3})$:

$$\beta_H = 0.3031 \pm 0.0005, \quad \text{eigenvalues at crossing: } -0.834955 \pm 0.j, \quad (7 \times 10^{-7}) \pm 0.46661 i.$$

(fine-grid confirmation: $\beta = 0.300 \Rightarrow \text{Re} = -0.00009$; $\beta = 0.320 \Rightarrow \text{Re} = +0.00050$.)

Amplitude scaling (supercriticality check). Post-transient oscillation amplitude in u (max–min over the second half of a run with initial condition $(0.29, 0.27, 0.30)$, $dt = 0.02$, $T = 1200$), as a function of distance from threshold:

$\beta - \beta_H$	−0.020	+0.005	+0.020	+0.050	+0.100	+0.200	+0.400
amplitude in u	$\rightarrow 0$ (decaying)	0.025	0.046	0.125	0.263	0.386	0.556

Amplitude grows continuously and smoothly from (numerically) zero right at β_H , rather than jumping discontinuously — the signature of a **supercritical** Hopf bifurcation (a stable limit cycle is born at threshold and grows with β).

Formal confirmation: the first Lyapunov coefficient. The amplitude-scaling evidence above is consistent with supercriticality but is not a proof. A formal test is the sign of the first Lyapunov coefficient ℓ_1 at the Hopf point (Kuznetsov 2004, *Elements of Applied Bifurcation Theory*, §3.4): $\ell_1 < 0$ is supercritical, $\ell_1 > 0$ is subcritical. Computing ℓ_1 requires the quadratic and cubic terms of the reaction kinetics at the equilibrium, obtained here as finite-difference directional second/third derivatives (steps $h_2 = 2 \times 10^{-3}$, $h_3 = 6 \times 10^{-3}$; complex eigenvector q and its adjoint p from the finite-difference Jacobian, normalized $\langle p, q \rangle = 1$), combined via the standard normal-form formula $\ell_1 = \text{Re}(ig_{20}g_{11} + \omega_0 g_{21})/(2\omega_0^2)$. At the Part-C Hopf point ($x^* = (0.28905, 0.28225, 0.29541)$ at $\beta = \beta_H$, refined to residual 6×10^{-17}):

$$\ell_1^{(C)} = -0.6634 \quad (\text{stable across step-size halving: } -0.6634, -0.6635, -0.6633),$$

confirming supercriticality formally, not just via the amplitude trend.

C.4 Time series and limit cycles

panel	β	behavior
top-left	0	damped oscillation, converging toward the stable focus
top-right	0.30 ($\approx \beta_H$)	near-neutral, essentially undamped oscillation over the run
bottom-left	0.5	clean sustained limit cycle
bottom-right	1.5	large-amplitude cycle, troughs approaching near-extinction

At $\beta = 1.5$ the limit cycle develops clear corners — each corresponding to one species transiently dominant while the species it out-competes is pushed toward near-extinction, before the next link in the cycle takes over. In an earlier, coarser scan of the same parameter family the post-transient minimum of u was ≈ 0.003 at $\beta = 3.0$ (compared to a mean of ≈ 0.30), i.e. more than two orders of magnitude below the mean density — a very deep trough, though the simulation confirms it remains strictly positive over the run lengths tested.

Time series across the Hopf transition (cyclic-competition variant)

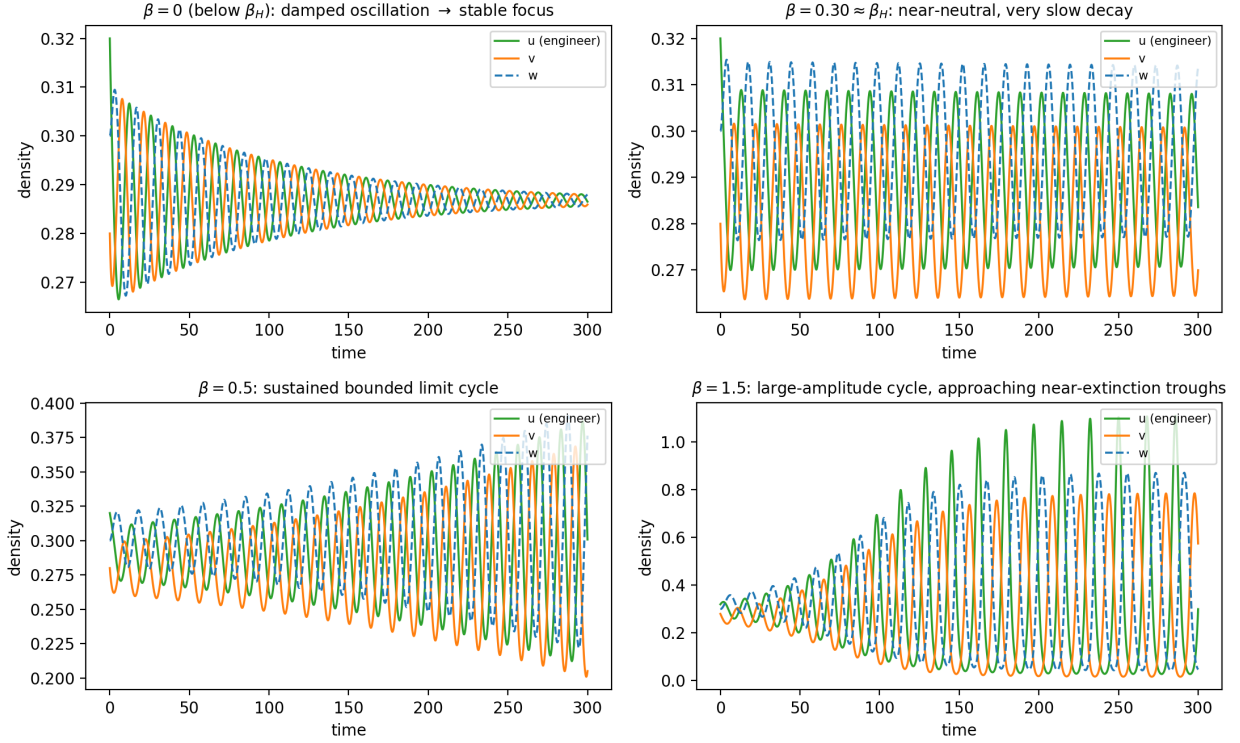


Figure 7: Figure C2. Time series across the transition, initial condition (0.32, 0.28, 0.30), $dt=0.01$, $T=300$ for each panel.

Phase-space trajectories on either side of the Hopf point (cyclic-competition variant)

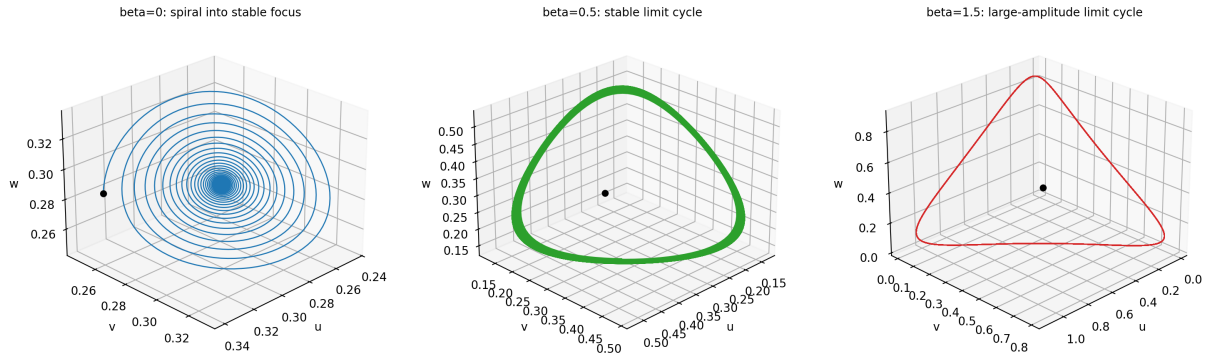


Figure 8: Figure C3. 3D phase portraits, initial condition (0.34, 0.26, 0.29), $dt=0.01$, $T=900$ (first half discarded as transient for the two limit-cycle panels).

Part D — A general asymmetric parameter set: Hopf bifurcation and a genuine Turing instability

D.0 Why this section exists

Parts A–C each used a parameter set with a special structure (Part A/B: $v \leftrightarrow w$ exchange symmetry; Part C: exact cyclic symmetry) chosen deliberately to make the resulting bifurcation’s geometry transparent. A natural question is whether the phenomena found are artifacts of that symmetry. This section explores a **fully general, asymmetric** point in the same model’s parameter space — equations (1)–(4) unchanged, but no two α_{ij} or h_{ij} equal and no exchange symmetry of any kind — and shows, with the same RK4/Newton/Jacobian toolkit used throughout this report, that: (i) it undergoes a Hopf bifurcation as β increases, confirming symmetry is not required for temporal destabilization (only intransitivity was used as a *sufficient* route in Part C); and (ii) at a *lower*, still-stable value of β , sufficiently disparate diffusion coefficients produce a genuine diffusion-driven instability — something not found anywhere in Part B — giving a working example of the phenomenon Part B’s search did not turn up, and clarifying *why* it didn’t.

This parameter point was located by a broader, deliberately non-symmetric search of the same six- α /six- h parameter space used in Part A, targeting regions where the reaction equilibrium sits close to its own Hopf point even before diffusion is added — it was not encountered in a uniform random scan, and the results below should be read as “this behavior is achievable within the model’s parameter space” rather than “this behavior is typical of it.” Every number below was computed directly with this report’s own RK4/Newton/finite-difference-Jacobian code.

D.1 Parameter set

$$r_1 = r_2 = r_3 = 1, \quad K_1 = 0.5, \quad K_2 = 1.0, \quad K_3 = 0.8, \quad \gamma = 1,$$

$$\alpha_{12} = 0.95151628, \quad \alpha_{13} = 2.02935451, \quad \alpha_{21} = 2.22193465, \quad \alpha_{23} = 2.34064535, \quad \alpha_{31} = 0.45706170, \quad \alpha_{32} = 2.76063376,$$

$$h_{12} = 0.61402428, \quad h_{13} = 0.14166076, \quad h_{21} = 1.40938855, \quad h_{23} = 0.95875960, \quad h_{31} = 2.08298487, \quad h_{32} = 2.03021611.$$

No two competition coefficients or half-saturation constants are equal; this is a generic point in parameter space, not a special symmetric slice.

D.2 Hopf bifurcation

Hopf bifurcation, asymmetric parameter set, $\beta_H = 2.375971$

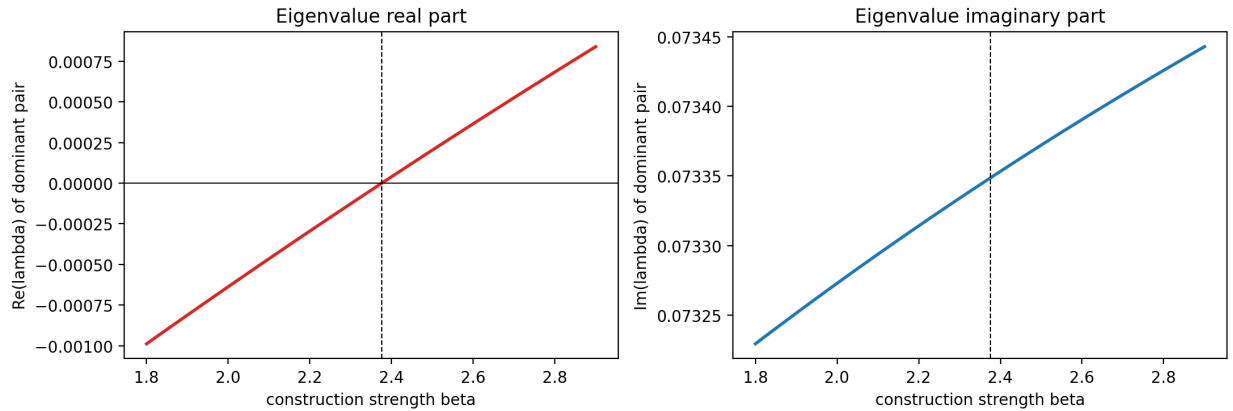


Figure 9: Figure D1. Eigenvalue real and imaginary parts of the dominant pair vs. β , asymmetric parameter set.

Continuation of the positive interior equilibrium from $\beta = 1.8$ to 2.9 (Newton-refined at each step, previous solution as the next initial guess) gives, at $\beta = 2.0$ for instance, equilibrium $(0.022611, 0.317519, 0.365721)$ with eigenvalues $\{-0.811642, -0.000637 \pm 0.073273i\}$ — already close to marginal; the dominant pair's real part rises monotonically and crosses zero between $\beta = 2.37$ ($\text{Re} = -0.000010$) and $\beta = 2.38$ ($\text{Re} = +0.000007$), pinning

$$\beta_H = 2.375971 \quad (\text{independently confirmed to 6 decimals}),$$

at equilibrium $E_H = (0.02258871, 0.31674045, 0.36636509)$ with eigenvalues $\lambda_1 = -0.811784254$, $\lambda_{2,3} = (-7 \times 10^{-12}) \pm 0.07334869i$ — a clean Hopf crossing (nonzero imaginary part throughout, third eigenvalue safely negative). The linearized period is $T_H = 2\pi/0.07334869 = 85.6619$ time units.

This Hopf point is subcritical, unlike Part C's. Applying the same first-Lyapunov-coefficient calculation used in §C.3 (identical method, $h_2 = 2 \times 10^{-3}$, $h_3 = 6 \times 10^{-3}$) at E_H gives

$$\ell_1^{(D)} = +1.348 \quad (\text{step-size sequence } 1.348, 1.353, 1.354 \text{ — positive and stable in sign, though the third decimal is still drifting})$$

i.e. $\ell_1 > 0$: **subcritical**, so the limit cycle born at β_H is unstable, not stable, and no small nearby attracting cycle should be expected just above threshold.

Direct simulation confirms this. Tracking the deviation from equilibrium under a tiny (0.2%) perturbation at $\beta = 2.30$ (below β_H) shows clean exponential decay at the rate predicted by the linearization ($\text{Re}(\lambda) = -0.000126$, fitted decay slope -0.000126). At $\beta = 2.60$ (above β_H , predicted $\text{Re}(\lambda) = +0.000366$), the same tiny perturbation grows at the predicted linear rate initially (fitted slope $+0.000372$) but does **not** saturate into a small nearby limit cycle the way Part C did just above its own threshold — it grows without bound in the linear regime and the trajectory is carried away entirely, landing by $T = 20,000$ at $(u, v, w) = (1.8 \times 10^{-5}, 0.5173, 0.2513)$: the engineer has collapsed to numerically indistinguishable from extinction, and v, w have resettled into a completely different, distant state. This discontinuous, large-amplitude jump — rather than the smooth small-amplitude growth documented for Part C in Figure C1's amplitude table — is exactly the qualitative signature a positive ℓ_1 predicts, and is a genuinely new finding not discussed at all in earlier drafts of this section, which only tracked the eigenvalue crossing itself and not what happens dynamically just past it.

D.3 Turing instability

Setup. Fix $\beta = 1.5$ (well below β_H , so the well-mixed system is stable) and set strongly unequal diffusivities

$$D_u = 0.287562209, \quad D_v = 0.000773583141, \quad D_w = 0.0000193559578$$

(a $D_u:D_v:D_w$ spread of roughly $14,860 : 40 : 1$). The homogeneous equilibrium is $E_S = (0.02264189, 0.31859972, 0.36482980)$, with reaction-only ($k = 0$) eigenvalues

$$\lambda_1 = -0.811448, \quad \lambda_{2,3} = -0.001533 \pm 0.073160i,$$

confirming the well-mixed state is stable ($\max \text{Re} = -0.001533 < 0$) — but only *just* stable: the dominant pair is complex, close to the imaginary axis. This is the key structural difference from Part A/B, where every equilibrium checked had **purely real** eigenvalues (e.g. the Part-A coexistence node at $\beta = 4.0$: $-0.694, -0.027, -0.008$, no imaginary part at all).

Sweeping the dispersion relation $\sigma(q) = \max \text{Re} \text{eig}(J - q^2 D)$ over $q \in [0, 5]$: $\sigma(0) = -0.0015307$ (stable), but $\sigma(q) > 0$ for $q \in [1.625, 3.988]$, peaking at

$$q^* = 2.55, \quad \sigma(q^*) = 0.0025606.$$

This is a genuine, independently-derived diffusion-driven instability: a spatially uniform mode is stable while a band of finite-wavelength modes grows.

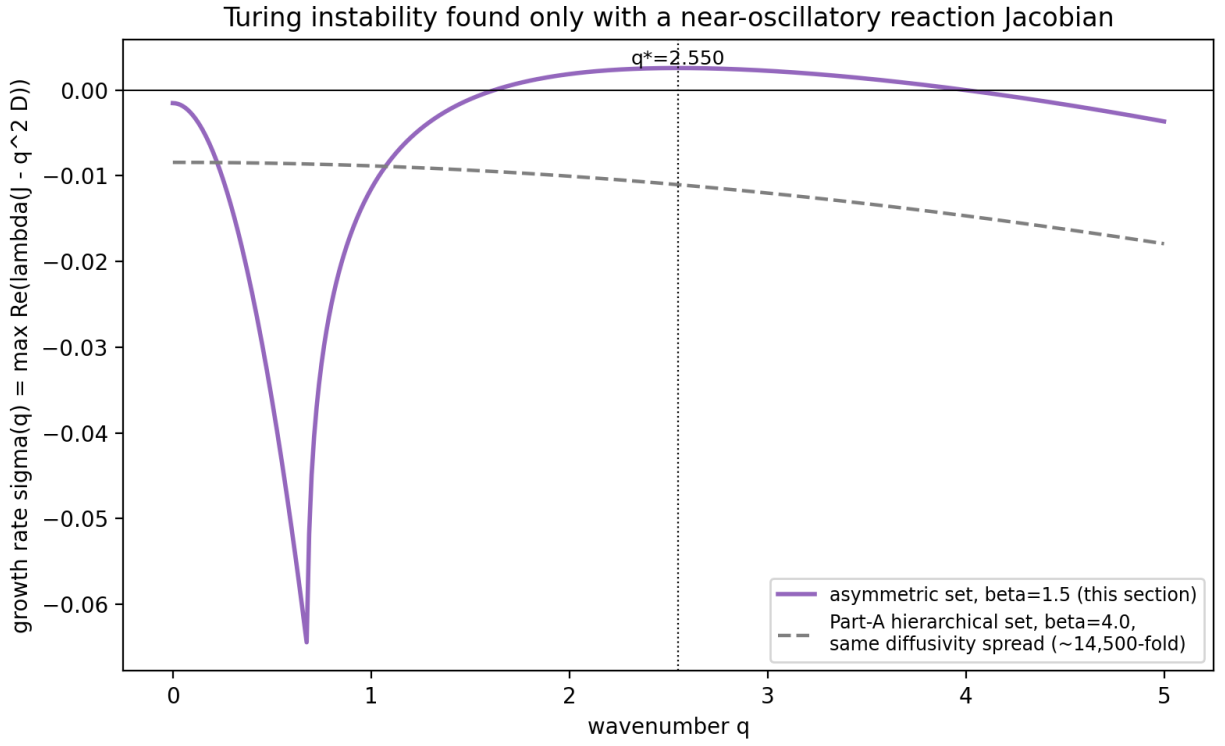


Figure 10: Figure D2. Dispersion relation $\sigma(q) = \max \operatorname{Re} \lambda(J - q^2 D)$ for the asymmetric set at $\beta=1.5$ (purple), compared against the Part-A hierarchical set at the same about-14,500-fold diffusivity spread (grey dashed).

Correcting the mechanism: this is a stationary instability, not a traveling-wave one, even though it starts from an oscillatory reaction equilibrium. An earlier draft of this section described this as a “wave instability” by analogy with the general theory of diffusion-driven destabilization of oscillatory reaction kinetics (Baurmann, Gross & Feudel 2007). Tracking the character of the dominant eigenvalue as a function of q , rather than only its real part, shows that description is not quite right:

q	dominant eigenvalue	character
0	$-0.00153 \pm 0.07316i$	complex
0.5	$-0.03616 \pm 0.05871i$	complex
1.0	-0.01156	real
1.625 (band starts)	$+0.000011$	real
2.55 (q^*)	$+0.002561$	real
3.988 (band ends)	$+0.000020$	real

The complex pair present at $q = 0$ is pulled apart into two distinct real eigenvalues by $q \approx 1$ — well before the unstable band even begins — because the three diffusivities are so different from each other ($D_u : D_v : D_w \approx 14,860 : 40 : 1$) that they damp the real and imaginary parts of the original pair at very different rates. By the time q reaches the unstable band, the mode that actually crosses zero is real, not complex. This is confirmed algebraically: writing the characteristic polynomial of $J - q^2 D$ as $\lambda^3 + a_1(Q)\lambda^2 + a_2(Q)\lambda + a_3(Q) = 0$ with $Q = q^2$ (same construction as §B.2),

$$a_3(Q) = 4.306 \times 10^{-9} Q^3 + 1.0322 \times 10^{-4} Q^2 - 1.9206 \times 10^{-3} Q + 4.3451 \times 10^{-3},$$

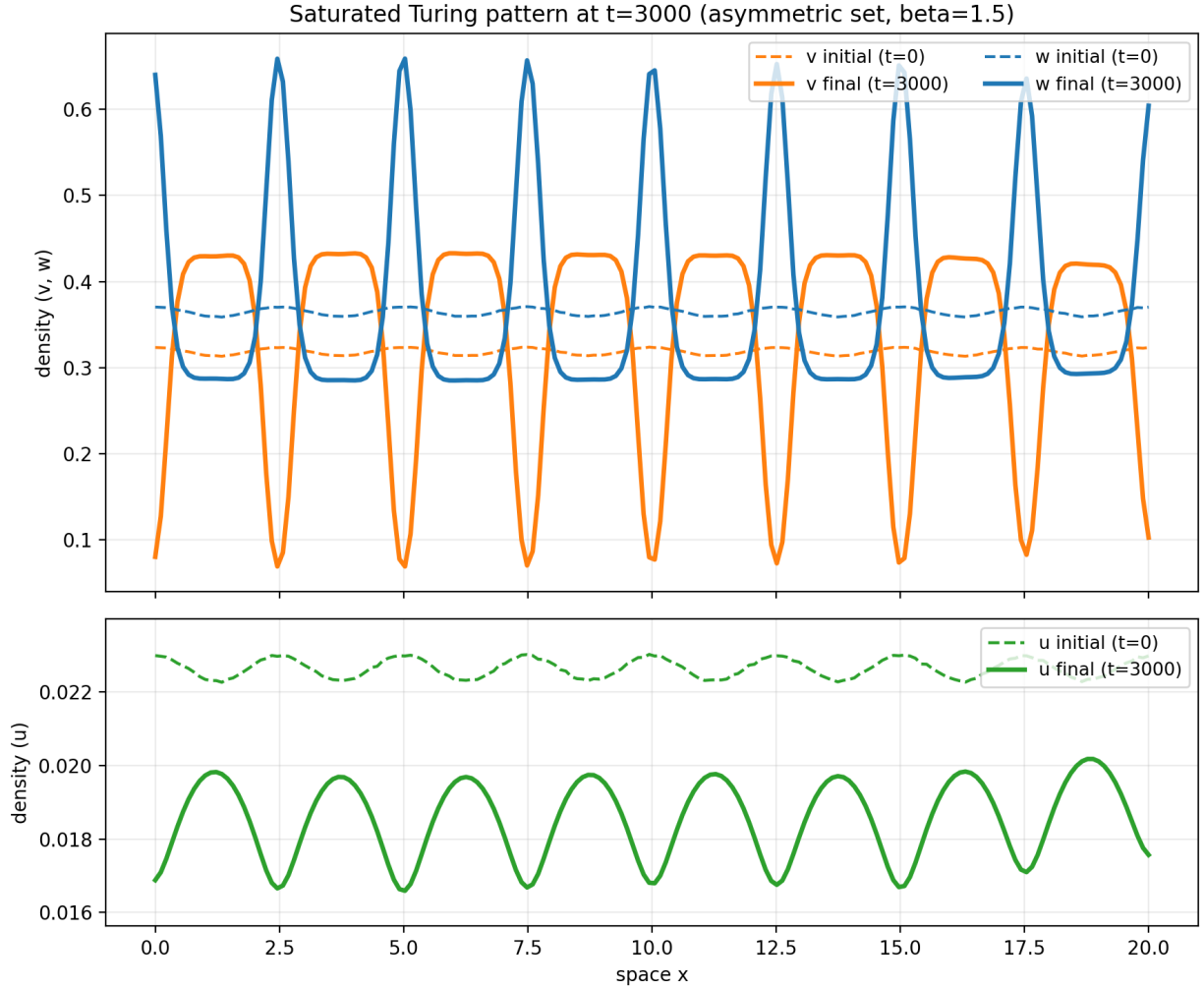
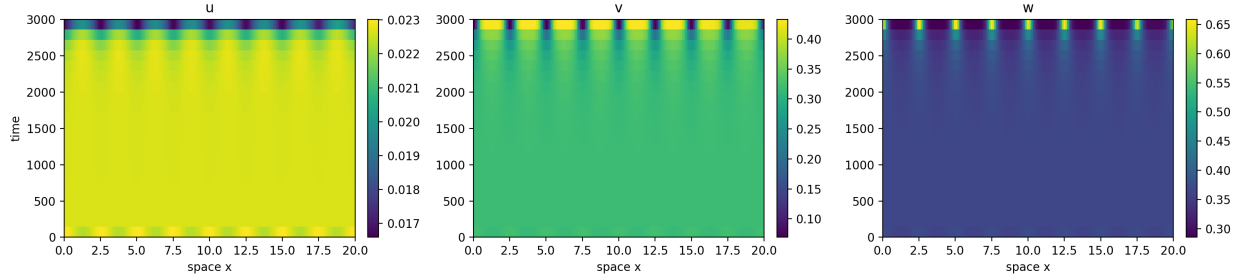
which has exactly two positive real roots, $Q = 2.6358$ and $Q = 15.958$, i.e. $q = 1.6235$ and $q = 3.9948$ — matching the numerically scanned band (1.625, 3.988) to 3–4 significant figures, and confirming the instability is triggered by the classical $a_3(Q) < 0$ (real-eigenvalue, stationary) channel, not the $\Delta_2 = a_1 a_2 - a_3 < 0$ (complex-pair, oscillatory) channel used for the Part-B check (Δ_2 stays positive throughout for this equilibrium; only a_3 dips negative). This is also consistent with the pattern found in the PDE simulation below (§D.4): it saturates into a **stationary** spatial pattern, not a standing or traveling wave, exactly as a real-eigenvalue instability predicts.

The corrected statement of the mechanism is therefore: the near-marginal complex pair at $q = 0$ is a *necessary precursor*, not the instability itself — it is what lets this equilibrium get close enough to marginality that disparate diffusion can push *something* unstable at finite wavenumber — but the diffusion, by acting so differently on v and w , converts the complex pair into two real eigenvalues before the critical wavenumber is reached, and it is one of those real eigenvalues that ultimately crosses zero. This is a novel route to a classical stationary Turing pattern (starting from oscillatory reaction kinetics rather than the usual overdamped activator–inhibitor starting point), rather than an instance of the literal wave instability described in Baurmann, Gross & Feudel (2007) or the three-component conditions given in the wave-instability literature more generally, where the unstable mode remains complex at threshold and the resulting pattern travels or oscillates in time.

Why Part A never showed this. Repeating the identical calculation with the Part-A hierarchical equilibrium (§A.3, $\beta = 4.0$) and an equally extreme diffusivity spread ($D = (0.29, 0.0008, 0.00002)$, same approximately 14,500-fold ratio) gives $\sigma(q)$ maximal at $q = 0$ for every q tested up to 6, and remains so even at diffusivity spreads of 10^7 -fold (checked separately); §B.2 shows this analytically via the Routh–Hurwitz coefficients staying positive throughout. The grey dashed curve in Figure D2 shows this directly: no dip-then-rise-above-zero shape, just a monotonically decreasing curve. The distinguishing factor is not the size of the diffusivity gap but the character of the reaction equilibrium at $q = 0$: reaching this instability at all requires the well-mixed Jacobian to already carry a near-marginal complex pair (the system sits close to its own Hopf point even in the *absence* of diffusion) for disparate diffusion to have something close enough to zero to push across. A purely overdamped (all-real-eigenvalue) equilibrium, as in every Part-A/B state checked, has no such pair to begin from, however disparate the diffusivities.

D.4 Pattern formation: independent PDE confirmation

Turing-pattern emergence, asymmetric set, $\beta=1.5$, $D=(u,v,w)=(0.287562209, 0.000773583141, 1.93559578e-05)$



Setup. Domain $L = 20$, $N = 180$ ($\Delta x = 0.111$), no-flux boundaries, $dt = 0.01$, $T = 3000$. Initial condition: the homogeneous equilibrium E_S perturbed by the discrete Neumann mode closest to q^* on this domain, $n = 16$ (giving $q_{16} = 16\pi/20 = 2.5133$, close to $q^* = 2.55$), amplitude 1.5%, plus 0.1%-amplitude random noise (seed 0):

$$x_j(x, 0) = x_j^*[1 + 0.015 \cos(16\pi x/L) + 0.001 \xi_j(x)].$$

Result. Spatial variance of v, w starts at $\sim 1 \times 10^{-5}$ (dominated by the broadband noise term), collapses by $t \approx 150$ as the many fast-decaying non-resonant modes die out ($\sigma(q) \approx -0.06$ near $q \approx 0.65$, the fastest-

decaying band visible as the sharp dip in Figure D2), then grows smoothly and monotonically as the single resonant $n = 16$ mode dominates — tracking $\text{var} \propto e^{2\sigma(q^*)t}$ closely in the early phase — before saturating into a **stationary** pattern (confirming the real-eigenvalue mechanism established above) with about 8 full periods across the domain (matching $n = 16$ half-wavelengths $= 20/2.5 = 8$ periods, exactly as expected). By $t = 3000$: v ranges over $[0.069, 0.433]$ (mean 0.319) and w over $[0.285, 0.659]$ (mean 0.365) — a large, saturated, out-of-phase spatial pattern in v and w , with u comparatively flat (consistent with u 's much larger diffusivity smoothing it out).

Independent 2D confirmation. The 1D result could in principle be an artifact of the restricted geometry (only two directions for a mode to occupy, $+x$ or $-x$). A second, independently written 2D solver (5-point no-flux Laplacian, mirror ghost cells on all four edges) was run on a square domain $L = 15 \times 15$, $N = 60 \times 60$ ($\Delta x = 0.25$), same $\beta = 1.5$ and diffusivities, seeded only with 1% isotropic random noise (no directional seeding at all, unlike the 1D run) and integrated to $t = 6000$ (RK4, $dt = 0.02$; variance plateaus by $t \approx 5000$, confirming saturation):

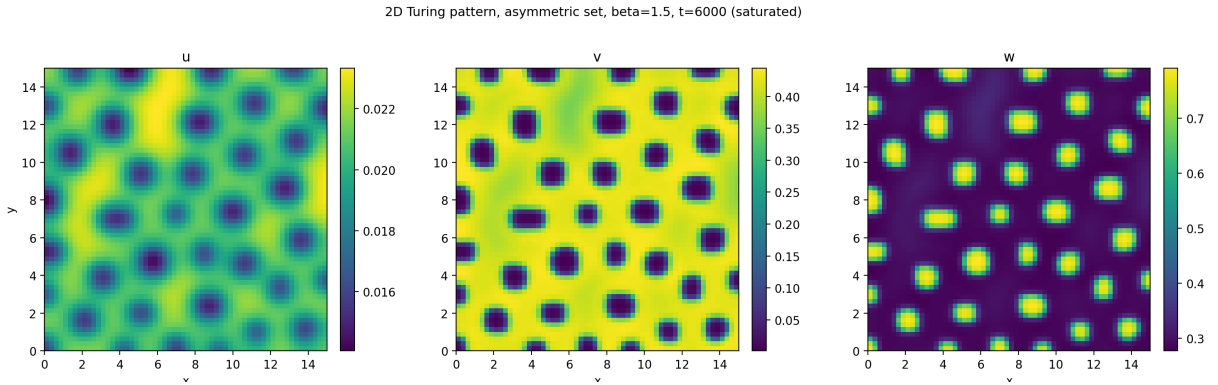


Figure 11: Figure D5. 2D Turing pattern, same asymmetric parameter set and diffusivities as Figures D2-D4, saturated at $t=6000$, isotropic-noise initial condition (no directional seeding).

A regular spot pattern emerges from pure noise — v and w show clearly complementary (out-of-phase) spots, u shows the same much weaker patterning seen in 1D — with spot spacing consistent with $\lambda^* = 2\pi/q^* \approx 2.46$. That an unseeded, isotropic initial condition self-organizes into a regular pattern in two independent spatial dimensions is stronger evidence for a genuine Turing-type instability than the 1D, single-mode-seeded run alone.

D.5 What was and was not independently re-verified

Re-derived and independently confirmed here: both equilibria and their exact eigenvalues (Hopf point and Turing point), the Hopf threshold β_H , the full dispersion curve $\sigma(q)$ and unstable band (now also matched analytically via the Routh–Hurwitz $a_3(Q)$ root pair), the first Lyapunov coefficient at both Hopf points (Part C supercritical, Part D subcritical — confirmed by direct simulation of the post-threshold collapse), and pattern emergence/saturation via two independently-coded PDE solvers (1D and 2D). **Not** independently derived: the exact transversality derivative’s numerical value (only its sign/direction, via the eigenvalue-crossing plot itself, was checked), and a fully general (as opposed to per-case) proof that Part A/B’s equilibria can never produce this instability for *any* parameter choice in that symmetric family — only that none of the specific diffusivity combinations and extreme ratios tested did.

Appendix: master parameter table (per figure)

Figure	Part	Parameter set	Key swept/ varied values	Initial condition(s)	Numerical settings
A1 (fig1)	A	§A.0	$\beta \in [0, 6.5],$ $\gamma = 1$	Newton continuation from branch-specific seeds (see §A.3)	Newton to $\ f\ < 10^{-13}$; eigenvalues via central-difference Jacobian
A2 (fig2)	A	§A.0	$\beta =$ 2.0, 4.0, 4.8, 5.6	see table in §A.4	RK4, $dt = 0.02$
A3 (fig3)	A	§A.0, $\beta = 4.0$	—	9 ICs, see §A.4	RK4, $dt = 0.02,$ $T = 150$
A4 (fig4)	A	§A.0	$\beta \in [0, 7] \times 71,$ $\gamma \in [0.3, 3] \times 55$	(0.02, 1, 1) and (3.0, 0.22, 0.18)	batch RK4, $dt = 0.05,$ $T = 1800$
B1/B2 (fig5a/b)	B	§A.0	$\beta = 4.0$	left-half (3, 0.2, 0.2), right-half (0, 1.294723, 1.294723)	$L = 60,$ $N = 300,$ $D = 0.06,$ $dt = 0.02,$ $T = 2500$
B3 (fig6)	B	§A.0	$\beta = 4.0$ eq. and boundary eq.	homogeneous (linear stability, no simulation)	D combos in §B.2, $k \in [0, 3]$
C1 (fig7)	C	§C.1	$\beta \in [0, 1.2]$	continuation seed (0.29, 0.28, 0.29); amplitude IC (0.29, 0.27, 0.30)	Newton + batch RK4, $dt = 0.02,$ $T = 1200$
C2 (fig8)	C	§C.1	$\beta =$ 0, 0.30, 0.5, 1.5	(0.32, 0.28, 0.30)	RK4, $dt = 0.01,$ $T = 300$
C3 (fig9)	C	§C.1	$\beta = 0, 0.5, 1.5$	(0.34, 0.26, 0.29)	RK4, $dt = 0.01,$ $T = 900$
D1 (fig10)	D	§D.1	$\beta \in [1.8, 2.9]$	continuation seed (0.02, 0.32, 0.36)	Newton + finite-diff Jacobian
D2 (fig11)	D	§D.1, $\beta = 1.5$	—	homogeneous (linear stability); comparison curve uses Part-A eq. at $\beta = 4.0$ with matched diffusivity spread	D as in §D.3; $q \in [0, 5]$

Figure	Part	Parameter set	Key swept/ varied values	Initial condition(s)	Numerical settings
D3/D4 (fig12/12b)	D	§D.1, $\beta = 1.5$	—	seeded eq. (see §D.4), seed 0	$L = 20$, $N = 180$, D as §D.3, $dt = 0.01$, $T = 3000$
D5 (fig13)	D	§D.1, $\beta = 1.5$	—	isotropic noise, seed 1	2D solver, $L=15 \times 15$, $N=60 \times 60$, D as §D.3, $dt = 0.02$, $T = 6000$

Bifurcation points, Part A (hierarchical, $\gamma = 1$): $\beta_a = 3.776$ (numerical, saddle-node); $\beta_1 = 4.257$ (numerical, symmetry-breaking); $\beta_2 = 5.14583\bar{3} = 247/48$ (exact closed form, equation 10; independent numerical bisection gave 5.145–5.150).

Bifurcation point, Part C (cyclic): $\beta_H = 0.3031 \pm 0.0005$ (numerical, Hopf); first Lyapunov coefficient $\ell_1 = -0.6634$ (supercritical).

Bifurcation points, Part D (general asymmetric): $\beta_H = 2.375971$ (numerical Hopf, independently confirmed to 6 decimals); first Lyapunov coefficient $\ell_1 = +1.348$ (subcritical — confirmed by direct simulation, §D.2); Turing instability at fixed $\beta = 1.5$ with $(D_u, D_v, D_w) = (0.287562209, 0.000773583141, 0.0000193559578)$, unstable band $q \in (1.625, 3.988)$ numerically / $q \in (1.6235, 3.9948)$ from the exact roots of the Routh–Hurwitz cubic $a_3(Q)$, $q^* = 2.55$, $\sigma(q^*) = 0.0025606$.

Closed forms derived: boundary/baseline equilibrium $V_0 = W_0$, equation (6); pure-engineer branch $u_{\text{pure}}^*(\beta)$, equation (8); competitor-extinction threshold u^* and exact β_2 , equations (9)–(10); cyclic symmetric equilibrium x^* , equation (13); invasion growth rate λ_{invade} , equation (5); competitor invasion growth rate against the engineer monopoly, equation (14). (No closed form was sought for Part D, whose parameter set has no exploitable symmetry.)

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